

Your Success is our Win (54)

$$A_{33} = (-1)^{2+3} \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix}$$

$$A_{33} = (-1)^4 (2-1)$$

$$A_{33} = 1$$

$$\text{adj of } A = \begin{bmatrix} 5 & -5 & -5 \\ -5 & 10 & 8 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

$$\text{adj of } A = \begin{bmatrix} 5 & 5 & 0 \\ -5 & 10 & 0 \\ -5 & 8 & 1 \end{bmatrix}_{3 \times 3}$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 5 & -5 & 0 \\ -5 & 10 & 0 \\ -5 & 8 & 1 \end{bmatrix}_{3 \times 3}$$

$$A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ -1 & 8/5 & 1/5 \end{bmatrix}_{3 \times 3}$$

Ex #2.4

① let $A = \begin{bmatrix} 2 & 5 & 0 \\ 3 & 4 & 6 \\ 1 & -5 & 1 \end{bmatrix}$, verify that $|A| = |A^T|$

Sol

Take L.H.S

$$= |A|$$

$$\Rightarrow A = \begin{bmatrix} 2 & 5 & 0 \\ 3 & 4 & 6 \\ 1 & -5 & 1 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow |A| = \begin{vmatrix} 2 & 5 & 0 \\ 3 & 4 & 6 \\ 1 & -5 & 1 \end{vmatrix}$$

$$\Rightarrow |A| = 2 \begin{vmatrix} 4 & 6 & -5 \\ 3 & 1 & 1 \end{vmatrix} - 5 \begin{vmatrix} 3 & 6 & 0 \\ 1 & 1 & -5 \end{vmatrix} + 0 \begin{vmatrix} 3 & 4 \\ 1 & -5 \end{vmatrix}$$

$$\Rightarrow |A| = 2(4+30) - 5(3-6) + 0(-15-4)$$

$$\Rightarrow |A| = 2(34) - 5(-3) + 0$$

$$\Rightarrow |A| = 68 + 15$$

$$\Rightarrow |A| = 83$$

Take R.H.S

$$= |A^T|$$

$$\Rightarrow A^T = \begin{bmatrix} 2 & 3 & 1 \\ 5 & 4 & 6 \\ 0 & -5 & 1 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^T = \begin{bmatrix} 2 & 3 & 1 \\ 5 & 4 & -5 \\ 0 & 6 & 1 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow |A^T| = \begin{vmatrix} 2 & 3 & 1 \\ 5 & 4 & -5 \\ 0 & 6 & 1 \end{vmatrix}$$

$$\Rightarrow |A^T| = 2 \begin{vmatrix} 4 & -5 \\ 6 & 1 \end{vmatrix} - 3 \begin{vmatrix} 5 & -5 \\ 0 & 1 \end{vmatrix} + 1 \begin{vmatrix} 5 & 4 \\ 0 & 6 \end{vmatrix}$$

Hence, proved!

② Without expanding, prove each of the following.

$$(i) \begin{vmatrix} x+y & y+z & z+x \\ 2 & x & y \\ 1 & 1 & 1 \end{vmatrix} = 0$$

Sol

By adding R_2 into R_1 , we get

$$\Rightarrow \begin{vmatrix} x+y+z & x+y+z & x+y+z \\ 2 & x & y \\ 1 & 1 & 1 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} x+y+z & 1 & 1 & 1 \\ 2 & x & y \\ 1 & 1 & 1 \end{vmatrix}$$

$\therefore$ two rows are identical

$\therefore$ Determinant = 0

$$\Rightarrow x+y+z(0) = 0$$

$$\Rightarrow 0 = 0$$

Hence, proved!

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(56)

$$(ii) \begin{vmatrix} 1 & x & y+z \\ 1 & y & z+x \\ 1 & z & x+y \end{vmatrix} = 0 \quad \begin{vmatrix} k & b & a & a+c+d \\ k & b & c & a+c+d \\ k & d & c & a+b+c \\ k & a & d & b+c+d \end{vmatrix}$$

Sol

By adding C_2 into C_3 ,
we get

$$\Rightarrow \begin{vmatrix} 1 & x & x+y+z \\ 1 & y & x+y+z \\ 1 & z & x+y+z \end{vmatrix}$$

$$\Rightarrow x+y+z \begin{vmatrix} 1 & x & 1 \\ 1 & y & 1 \\ 1 & z & 1 \end{vmatrix}$$

"two rows are identical from C_1 and $a+b+c+d$
 $\therefore$ Determinant = 0

$$\Rightarrow x+y+z(0)$$

$$\Rightarrow 0 = 0$$

proved!

$$(iii) \begin{vmatrix} k & b & a & c+d \\ k & b & c & a+d \\ k & d & c & b+a \\ k & a & d & b+c \end{vmatrix} = 0$$

Sol

By adding C_3 to C_4 ,
we get.

By adding C_2 into C_4 ,
we get,

$$\begin{vmatrix} k & b & a & a+b+c+d \\ k & b & c & a+b+c+d \\ k & d & c & a+b+c+d \\ k & a & d & a+b+c+d \end{vmatrix}$$

Taking k as common

from C_1 and $a+b+c+d$
from C_4 .

$$= k(a+b+c+d) \begin{vmatrix} 1 & b & a & 1 \\ 1 & b & c & 1 \\ 1 & d & c & 1 \\ 1 & a & d & 1 \end{vmatrix}$$

"two columns are identical"
 $\therefore$ determinant = 0

$$\Rightarrow k(a+b+c+d)(0)$$

$$= 0$$

$$= R.H.S$$

proved!

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① without expanding determinants prove that

$$(i) \begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} = 0$$

Sol

By adding C_2 into C_3 ,
we get,

$$= \begin{vmatrix} 1 & a & a+b+c \\ 1 & b & a+b+c \\ 1 & c & a+b+c \end{vmatrix}$$

Taking $a+b+c$ as common
from C_3

$$= a+b+c \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix}$$

"two columns are identical"
 $\therefore$ determinant = 0

$$= a+b+c(0)$$

$$= 0$$

$$= R.H.S$$

proved!

$$(ii) \begin{vmatrix} x+1 & x+2 & x+3 \\ x+4 & x+6 & x+7 \\ x+7 & x+9 & x+n \end{vmatrix} = 0$$

Sol

Take L.H.S

$$= \begin{vmatrix} x+1 & x+2 & x+3 \\ x+4 & x+6 & x+7 \\ x+7 & x+9 & x+n \end{vmatrix}$$

$$= \begin{vmatrix} x+1 & x+2 & x+3 \\ 3 & 3 & 3 \\ x+7 & x+9 & x+1 \end{vmatrix} \quad R_2 - R_1 \rightarrow R_2$$

$$= \begin{vmatrix} x+1 & x+2 & x+3 \\ 3 & 3 & 3 \\ 6 & 6 & 6 \end{vmatrix} \quad R_3 - R_1 \rightarrow R_3$$

Taking 3 as common from R_2

$$= 3 \times 6 \begin{vmatrix} x+1 & x+2 & x+3 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

"two rows are identical"
 $\therefore$ determinant = 0

$$= 18(0)$$

$$= 0$$

$$= R.H.S \text{ proved!}$$

57a

④ Without expanding the determinant, prove that

$$\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix} = \begin{vmatrix} x & y & z \\ p & a & r \\ a & b & c \end{vmatrix}$$

S1

Taking 1st R.H.S

$$= \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$$

$$\therefore |D| = |D^T|$$

$$\therefore \begin{vmatrix} y & x & z \\ b & a & c \\ q & p & r \end{vmatrix}$$

$$= \begin{vmatrix} x & y & z \\ a & b & c \\ p & q & r \end{vmatrix} C_{12}$$

$$= \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} R_{12}$$

$$L.H.S = 1^{st} R.H.S$$

Taking 2nd R.H.S

$$= \begin{vmatrix} x & y & z \\ p & a & r \\ a & b & c \end{vmatrix}$$

$$= \begin{vmatrix} x & y & z \\ p & a & r \\ a & b & c \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} R_{12}$$

$$L.H.S = 2^{nd} R.H.S$$

Hence,

$$L.H.S = 1^{st} R.H.S = 2^{nd} R.H.S$$

57b

⑤ If a, b & c are different and $\Delta = \begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$

then show that $1+abc = 0$

Sj

let,

$$\Delta = \begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} + \begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix}$$

$$\Delta = \Delta_1 + \Delta_2 \quad \text{--- ①}$$

Take Δ_1 :

$$\Delta_1 = \begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix} \quad \text{Taking } abc \text{ common from}$$

$$\Delta_1 = abc \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \quad \text{Taking } a, b, c \text{ Common from } R_1, R_2, R_3 \text{ respectively.}$$

$$\Delta_1 = -abc \begin{vmatrix} a & 1 & a^2 \\ b & 1 & b^2 \\ c & 1 & c^2 \end{vmatrix} C_{12}$$

$$\Delta_1 = abc \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} C_{13}$$

$$\therefore \Delta_1 = \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix}$$

$$\therefore \Delta_1 = abc \Delta_2$$

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Put value of Δ_2 ①

$$\Delta = \Delta_1 + abc \Delta_1$$

$$\therefore \Delta = 0$$

$$0 = \Delta_1 + abc \Delta_1$$

$$\Delta_1(1+abc) = 0$$

$$1+abc = 0$$

$$\Delta_1$$

$$1+abc = 0$$

proved!

Ex#2.5

① Reduce the following matrices into echelon form using elementary row operation.

$$(i) \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix}$$

Sol

$$\approx \begin{bmatrix} -1 & -5 & -3 \\ -3 & 5 & 6 \\ 5 & 9 & 3 \end{bmatrix} R_{1 \leftrightarrow 3}$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ -3 & 5 & 6 \\ 5 & 9 & 3 \end{bmatrix} R_1 \times (-1)$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ -3+3(1) & 5+3(5) & 6+3(3) \\ 5-5(1) & 9-5(5) & 3-5(3) \end{bmatrix} R_2 + 3R_1, R_3 - 5R_1$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ 0 & 20 & 15 \\ 0 & -16 & -12 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ 0 & (20)(\frac{1}{20}) & (15)(\frac{1}{20}) \\ 0 & -16 & -12 \end{bmatrix} (\frac{1}{20})R_2$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & \frac{3}{4} \\ 0 & -16 & -12 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & \frac{3}{4} \\ 0 & -16+16 & -12+16(\frac{3}{4}) \end{bmatrix} 16R_3 + R_2$$

$$\approx \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & \frac{3}{4} \\ 0 & 0 & 0 \end{bmatrix}$$

It is an echelon form of given matrix.

$$(ii) \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{bmatrix}$$

Sol

$$\approx \begin{bmatrix} 1 & 2 & -1 \\ 2-2 & 4-2(2) & 1-2(-1) \\ 3-3 & 6-3(2) & 2-3(-1) \end{bmatrix} R_2 - 2R_1, R_3 - 3R_1$$

$$\approx \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} R_3 - R_2$$

It is an echelon form of given matrix.